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Indium(III) Triflate-Catalyzed Efficient Prins-Type Cyclization of β-Allenols and Aldehydes

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Abstract: We report here an indium(III) triflate ($In(OTf)_3$)-catalyzed Prins-type cyclization of disubstituted β-allenols in the presence of aldehydes in dichloromethane (CH_2Cl_2) affording different 3,6-dihydro-2*H*-pyran at room temperature or 45 °C. The high efficiency of the reaction and easily available nature of the starting materials/catalyst make it useful in synthesizing different dihydropyran derivatives. A rationale for the formation of all the products was proposed.

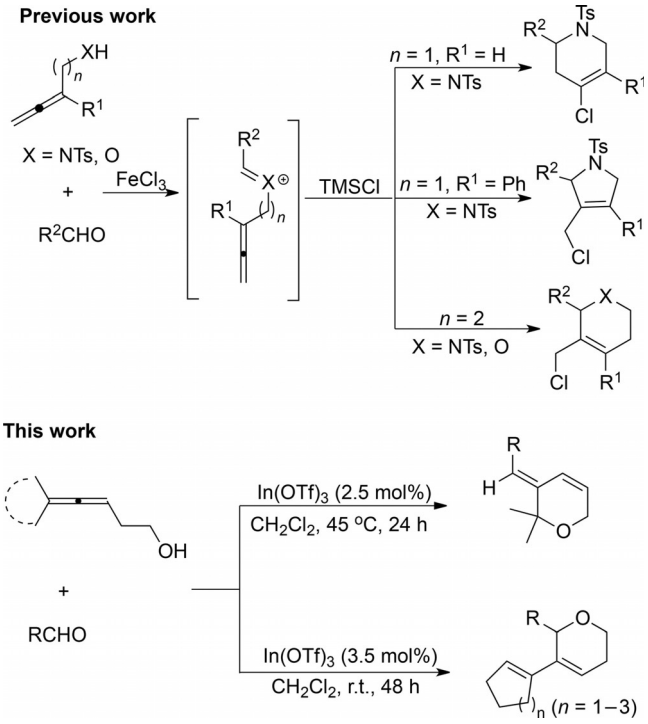
Keywords: β-allenols; aldehydes; indium(III) triflate; Prins reaction; 2*H*-pyrans

Introduction

Dihydropyran skeletons have been found in many natural products.[1] Many methodologies have been developed for the synthesis of dihydropyran derivatives, including hetero-Diels–Alder[2] and olefin metathesis reactions.[3] The Prins cyclization is a highly efficient approach toward heterocycles under mild reaction conditions.[4,5] Recently, our group has reported the $FeCl_3$-catalyzed cyclization of terminal α- or β-sulfonamidoallenes or β-allenenols and aldehydes in the presence of TMSCl to form 4-chloro-1,2,3,6-tetrahydropyridines, 3-chloromethyl-2,5-dihydro-1*H*-pyrroles, or 1,2,5,6-tetrahydro-1*H*-pyridines or 5,6-dihydro-2*H*-pyrans (Scheme 1).[6] Herein, we report a highly efficient $In(OTf)_3$-catalyzed Prins-type cyclization of 5,5-disubstituted 3,4-allenols and aldehydes to synthesize 3,6-dihydro-2*H*-pyrans or 5,6-dihydro-2*H*-pyrans (Scheme 1).

Results and Discussion

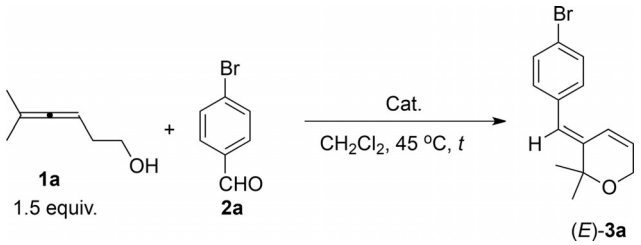
Our initial work began with a 5,5-disubstituted-3,4-allenol, i.e., 5-methylhexan-3,4-dienol **1a** (0.15 mmol, 1.5 equiv.), and 4-bromobenzaldehyde **2a** (0.1 mmol, 1.0 equiv.). Interestingly, the reaction with or without 1.0 equiv. of TMSCl under the catalysis of $FeCl_3$[5] (30 mol%) in CH_2Cl_2 at 45 °C failed to afford a 4-chloro-1,2,3,6-tetrahydropyridine-type product. Instead, a new cyclization product (*E*)-**3a** was formed in 43% yield after 12 h (Table 1, entry 1). In the absence of TMSCl, the yield was improved to 63% (Table 1, entry 2). When we used $InCl_3$[7] as the catalyst, the



Scheme 1. Cyclization of β-allnenols and aldehydes.

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Table 1. The optimization of the reaction conditions.[a]



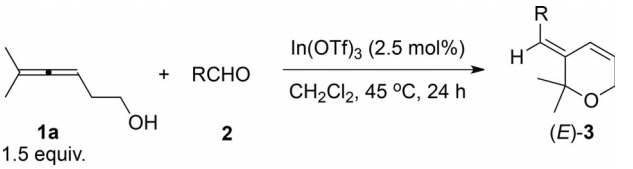
Entry	Cat. (mol%)	*t* [h]	Yield of (*E*)-**3a** [%][b]
1[c]	$FeCl_3$ (30)	12	43
2	$FeCl_3$ (30)	12	63
3	$InCl_3$ (30)	12	58
3	$In(OTf)_3$ (30)	12	99
5	$In(OTf)_3$ (2.5)	24	100
6[d]	$In(OTf)_3$ (2.5)	24	82
7[e]	TfOH (2.5)	24.8	54
8[f]	–	24	–
9[g]	$In(OTf)_3$ (2.5)	24	86

[a] The reaction was conducted using **1a** (0.15 mmol), **2a** (0.1 mmol), and CH_2Cl_2 (1 mL) at 45 °C.
[b] Determined by ^1H NMR analysis with mesitylene as the internal standard.
[c] 1.0 equiv. of TMSCl was used.
[d] 1.2 equiv. of **1a** were used and 2% of **2a** was recovered.
[e] 4 mL of CH_2Cl_2 were used.
[f] 62% of **2a** was recovered.
[g] The reaction was conducted under an argon atmosphere.

yield dropped slightly (Table 1, entry 3). Excitingly, when $In(OTf)_3$ was used instead of $InCl_3$, the yield was improved to 99% (Table 1, entry 4)! Moreover, the loading of $In(OTf)_3$ can be reduced to 2.5 mol% and a quantitative yield was observed after 24 h (Table 1, entry 5). Reducing the amount of **1a** led to a lower yield of the product (Table 1, entry 6). The efficiency of the catalyst depends on the Lewis acidity. As a control experiment, the cyclization product (*E*)-**3a** was obtained in 54% yield when TfOH (2.5 mol%) was used as the catalyst (Table 1, entry 7). No product was obtained in the absence of a catalyst (Table 1, entry 8). The yield dropped to 86% when the reaction was conducted in an argon atmosphere (Table 1, entry 9). Thus, **1a** (1.5 equiv.), **2a** (0.1 mmol), and $In(OTf)_3$ (2.5 mol%) in CH_2Cl_2 at 45 °C were defined as the optimized reaction conditions for further study.

With the optimal reaction conditions in hand, we then investigated the scope of the aldehydes. Aromatic aldehydes substituted by electron-withdrawing groups, such as *p*-Br, *p*-Cl and *p*-MeOOC, afforded (*E*)-**3a**–**3c** in good yields (Table 2, entries 1–3). The C=C bond configuration in the product was further established by an X-ray diffraction study of (*E*)-**3c** (Figure 1).[8] Aromatic aldehydes substituted with *p*-MeO, *p*-Me and *p*-Ph groups could also react with 5-methylhexan-3,4-dienol **1a** to form the corresponding

Table 2. The scope of aldehydes.[a]



Entry	R	Yield of (*E*)-**3** [%][b]
1	*p*-BrC_6H_4 (**2a**)	89 (*E*)-**3a**
2	*p*-ClC_6H_4 (**2b**)	85 (*E*)-**3b**
3	*p*-$MeO_2CC_6H_4$ (**2c**)	67 (*E*)-**3c**
4	*p*-$MeOC_6H_4$ (**2d**)	76 (*E*)-**3d**
5	*p*-MeC_6H_4 (**2e**)	83 (*E*)-**3e**
6	*p*- PhC_6H_4 (**2f**)	68 (*E*)-**3f**
7	*m*-$MeOC_6H_4$ (**2g**)	77 (*E*)-**3g**
8	*o*-ClC_6H_4 (**2h**)	72 (*E*)-**3h**
9	*o*-$MeOC_6H_4$ (**2i**)	76 (*E*)-**3i**
10	cyclohexyl (**2j**)	48 (*E*)-**3j**

[a] The reaction was conducted using **1a** (0.6 mmol), aldehyde (0.4 mmol), $In(OTf)_3$ (2.5 mol%) in CH_2Cl_2 (4 mL) at 45 °C.
[b] Isolated yield.

cyclization products in 68–83% yields (Table 2, entries 4–6). What is more, *ortho*- and *meta*-substituted aromatic aldehydes may also be applied in this reaction (Table 2, entries 7–9). Cyclohexanecarbaldehyde **2j** can afford (*E*)-**3j** with 48% yield (Table 2, entry 10). However, the reaction became complicated when other aliphatic aldehydes such as *n*-butyraldehyde and *n*-pentanal were employed.

Based on these results and our recent reports on the Prins reactions,[6] a plausible mechanism for this reaction shown in Scheme 2 was proposed: With the catalysis of $In(OTf)_3$, a coordination complex between $In(OTf)_3$ and aldehyde **2** was formed, which was attacked by alcohol unit to generate **Int 1**. Oxonium intermediate **Int 2** was then formed, which underwent cyclic nucleophilic attack by the 3-position C=C bond of the allene unit to form cyclic cationic intermediate **Int 3**. Intermediate product **4** would be obtained after β-H^+ elimination of **Int 3** forming an extra C=C bond in the six-membered ring with R^2 being Me. In the presence of $In(OTf)_3$, the cleavage of the C–O bond would occur to generate allylic **Int 4**. Due to the presence of the two methyl groups, the cationic center was delocalized and re-localized to afford a new more stable allylic **Int 5**. Cyclic C–O bond formation would form the product **3** (Scheme 2). The stereoselectivity of the C=C bond in the product **3** was determined by the possible interaction of the R^1 and the two methyl groups.

In fact, when we used the electron-deficient 4-nitrobenzaldehyde **2k** as the starting material, two such products 2-(4-nitrophenyl)-3-(propan-2-ylidene)-3,6-dihydro-2*H*-pyran **4a** and 6-(4-nitrophenyl)-5-(prop-1-

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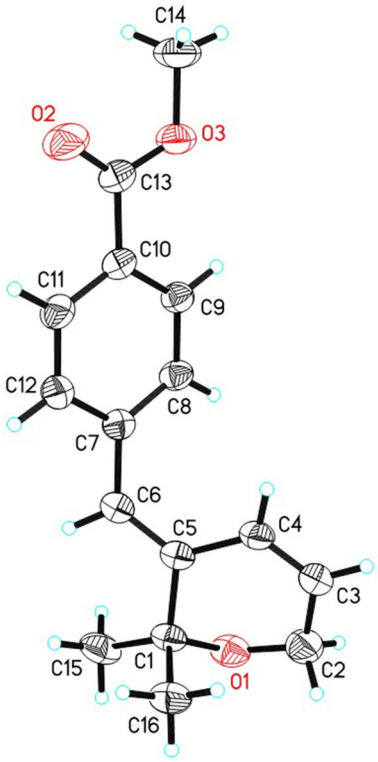
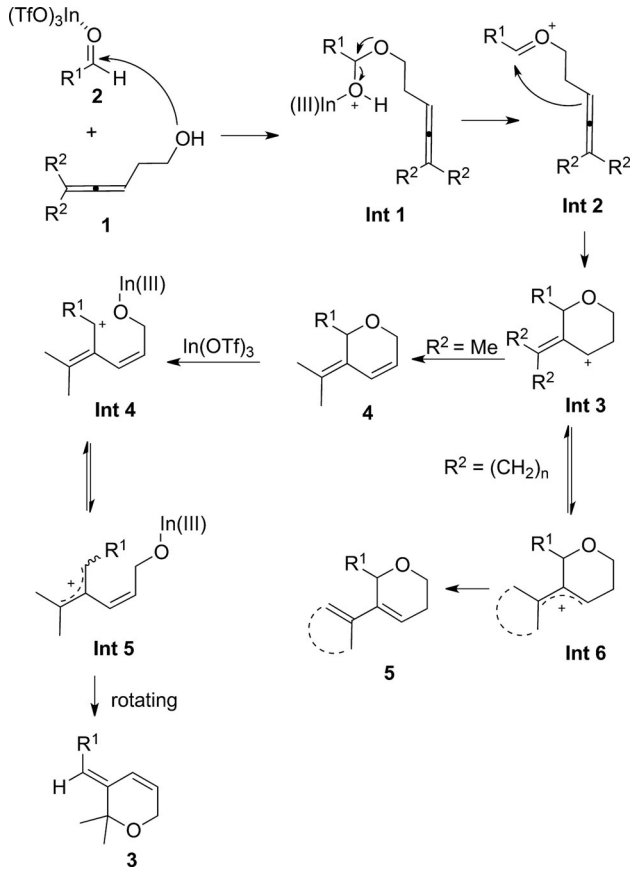
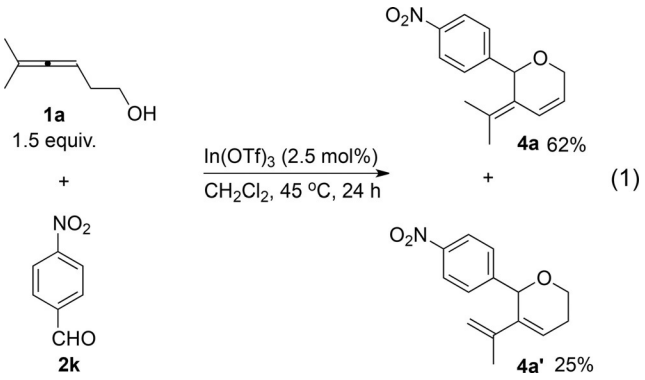


Figure 1. X-ray crystal structure of (*E*)-**3c**.

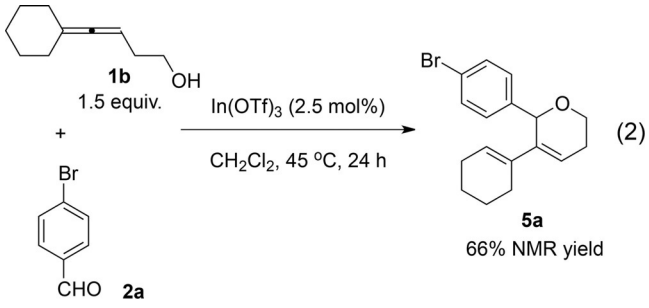


Scheme 2. A proposed mechanism for the cyclization of β-allenols and aldehydes.

en-2-yl)-3,6-dihydro-2*H*-pyran **4a'** were obtained in 62% and 25% yields, respectively [Eq. (1)].



When 5,5-pentamethylene-3,4-pentadienol **1b** and 4-bromobenzaldehyde **2a** were reacted under the standard conditions, **5a** was formed in 66% NMR yield, exclusively, most probably due to the favorable formation of two cyclic C=C bonds [Eq. (2)].

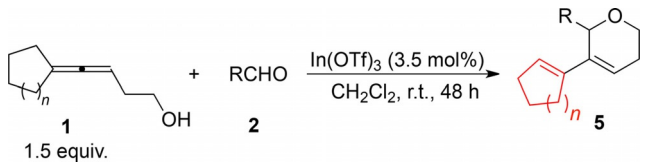


After numerous trials, we found that 5,5-pentamethylene-3,4-pentadienol **1b** (1.5 equiv.) and 4-bromobenzaldehyde **2a** (0.2 mmol) could react in the presence of $In(OTf)_3$ (3.5 mol%) even at room temperature for 48 h to form product **5a** in 80% isolated yield exclusively (Table 3, entry 1). Aromatic aldehydes substituted with *p*-Cl, *p*-MeOOC, *p*-MeO, *p*-Ph and *m*-MeO all afforded good yields (Table 3, entries 2–5 and 7). What is more, 4-nitrobenzaldehyde **2k** could also react to afford **5f** in 40% yield (Table 3, entry 6). Cyclohexanecarbaldehyde **2j** could be applied in this Prins cyclization to afford **5h** in 74% yield (Table 3, entry 8). *n*-Butyraldehyde and *n*-pentanal may also react with **1b** to form **5i** and **5j** in 53% and 55% yield within 12 h, respectively (Table 3, entries 9 and 10). We also examined the scope of the β-allenols.[9] 5,5-Tetramethylene-3,4-pentadienol **1c** and 5,5-hexamethylene-3,4-pentadienol **1d** could also react with 4-bromobenzaldehyde **2a** to afford the corresponding products **5k** and **5l** in moderate yields (Table 3, entries 11 and 12). The structure of the product **5** was established by the X-ray diffraction study of **5c** (Figure 2).[10]

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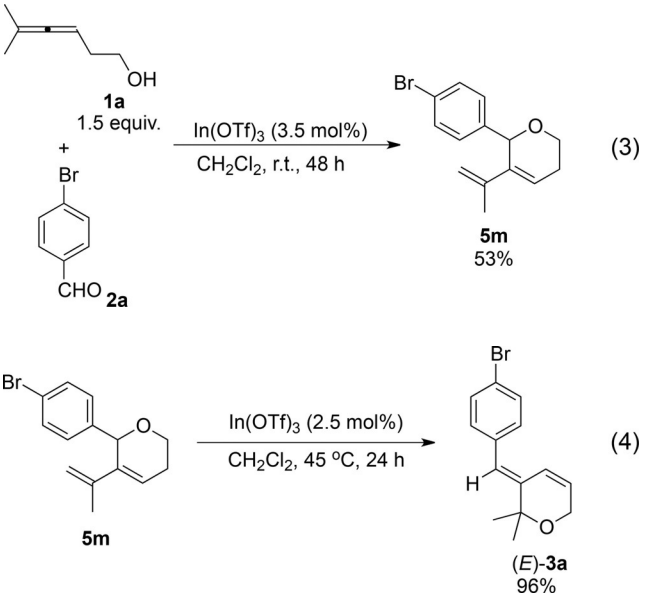
Table 3. The scope of the substrates.[a]



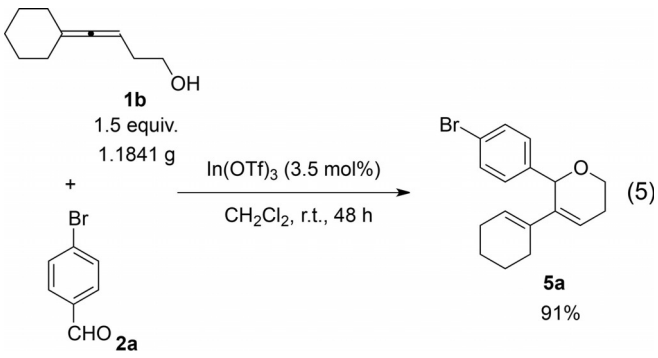
Entry	n	R	Yield of **5** [%][b]
1	2 (**1b**)	*p*-BrC_6H_4 (**2a**)	80 (**5a**)
2	2 (**1b**)	*p*-ClC_6H_4 (**2b**)	75 (**5b**)
3	2 (**1b**)	*p*-$MeO_2CC_6H_4$ (**2c**)	83 (**5c**)
4	2 (**1b**)	*p*-$MeOC_6H_4$ (**2d**)	88 (**5d**)
5	2 (**1b**)	*p*-PhC_6H_4 (**2f**)	69 (**5e**)
6	2 (**1b**)	*p*-$NO_2C_6H_4$ (**2k**)	40 (**5f**)
7	2 (**1b**)	*m*-$MeOC_6H_4$ (**2g**)	85 (**5g**)
8[c]	2 (**1b**)	cyclohexyl (**2j**)	74 (**5h**)
9[d,e]	2 (**1b**)	C_3H_7 (**2k**)	53 (**5i**)
10[d,f]	2 (**1b**)	C_4H_9 (**2l**)	55 (**5j**)
11[c]	1 (**1c**)	*p*-BrC_6H_4 (**2a**)	65 (**5k**)
12[c]	3 (**1d**)	*p*-BrC_6H_4 (**2a**)	65 (**5l**)

[a] The reaction was conducted using **1b** (0.6 mmol), aldehyde (0.4 mmol), $In(OTf)_3$ (3.5 mol%) in CH_2Cl_2 (4 mL) at room temperature.
[b] Isolated yield.
[c] The reaction was conducted with 1 mmol scale.
[d] The reaction time was 12 h.
[e] 2% of **2k** were recovered by NMR.
[f] 6% of **2l** were recovered by NMR.

As demonstrated in Scheme 2, the formation of **3** should require a high activation barrier and the formation of **5** *via* **Int 6** should be favored. In fact, when 5-methylhexa-3,4-dienol **1a** was treated with 4-bromobenzaldehyde **2a** at room temperature, the corresponding product **5m** was afforded exclusively [compare Eq. (3) with entry 1 in Table 2], indicating that the reaction temperature may determine the structures of the products. In fact, it was observed that **5m** was converted to (*E*)**-3a** in 96% yield under the catalysis of $In(OTf)_3$ at 45 °C for 24 h [Eq. (4)].



The reaction can be easily conducted on a 1 g scale to afford **5a** in 91% yield [Eq. (5)].



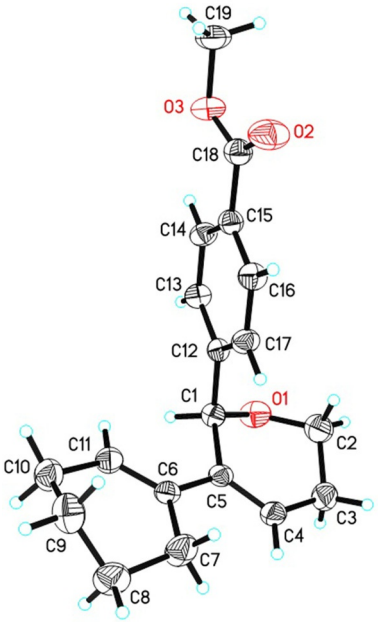


Figure 2. X-ray crystal structure of **5c**.

Conclusions

In conclusion, we have developed an efficient $In(OTf)_3$-catalyzed Prins-type cyclization of β-allenols and aldehydes affording 3,6-dihydro-2*H*-pyrans or 5,6-dihydro-2*H*-pyrans depending on the reaction conditions. Mechanistic studies confirmed the conversion between these two classes of products leading to a rationale for their formation. Due to the easily availability of the starting materials,[9a,11] the high efficiency of the catalyst and the simple operation, the reaction should be useful in organic synthesis. Further

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studies on an asymmetric version of this reaction and the use of β-sulfonamidoallenes as starting materials are being conducted in our laboratory.

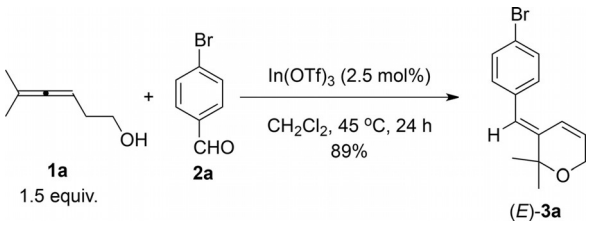
Experimental Section

General Information

All reactions were carried out in oven-dried Schlenk tubes unless otherwise noted. $In(OTf)_3$ (99.5%) was purchased from Aladdin and the petroleum ether (boiling range: 60–90 °C) was purchased from Adamas. CH_2Cl_2 was dried with CaH_2 and distilled freshly before use. Aliphatic aldehydes were distilled freshly before use. Other reagents were used without further treatment. All the temperatures are referred to the oil baths used.

Experimental Details and Analytical Data

Typical Procedure I: (*E*)-3-(4-Bromobenzylidene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3a]:

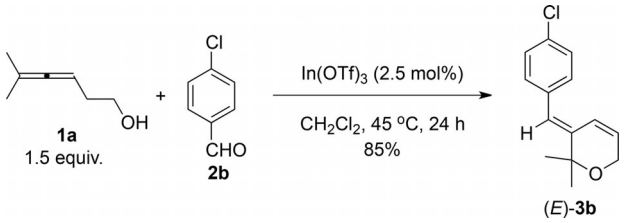


To a Schlenk tube were added $In(OTf)_3$ (5.5 mg, 0.01 mmol), 4-bromobenzaldehyde **2a** (73.9 mg, 0.4 mmol), 5-methylhexan-3,4-dienol **1a** (68.0 mg, 0.6 mmol), and 4 mL of CH_2Cl_2 sequentially. Then the Schlenk tube was placed in a pre-heated oil bath at 45 °C with stirring for 24 h. The reaction was complete as monitored by TLC. Then 10 mL of CH_2Cl_2 and 10 mL of ethyl ether were used to transfer the reaction mixture. After evaporation, 28 μL of mesitylene were added to the residue for the ^1H NMR analysis and then purified by chromatography on silica gel to afford (*E*)-**3a** (eluent: *n*-hexane/ethyl acetate = 45/1) as a light yellow solid; yield: 99.5 mg (89%); mp 49–52 °C (*n*-hexane/CH_2Cl_2). ^1H NMR (400 MHz, $CDCl_3$): $\delta = 7.44$ (d, $J = 8.4$ Hz, 2H, Ar-H), 7.13 (d, $J = 8.0$ Hz, 2H, Ar-H), 6.53–6.48 (m, 1H, =CH), 6.34 (s, 1H, =CH), 5.96–5.91 (m, 1H, =CH), 4.31–4.28 (m, 2H, OCH_2), 1.44 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 140.2$, 135.8, 131.2, 131.0, 128.8, 121.4, 121.2, 120.6, 73.6, 61.1, 25.9; MS (EI): *m/z* (%) = 280 [M^+(^{81}Br), 25.91], 278 [M^+(^{79}Br), 25.13], 43 (100); IR (neat): $\nu = 3047$, 2973, 2930, 2815, 1636, 1583, 1482, 1441, 1414, 1376, 1359, 1305, 1280, 1258, 1239, 1218, 1197, 1179, 1146, 1122, 1103, 1067, 1007 cm^{-1}; anal. calcd. for $C_{14}H_{15}BrO$: C 60.23, H 5.42; found: C 60.30, H 5.41.

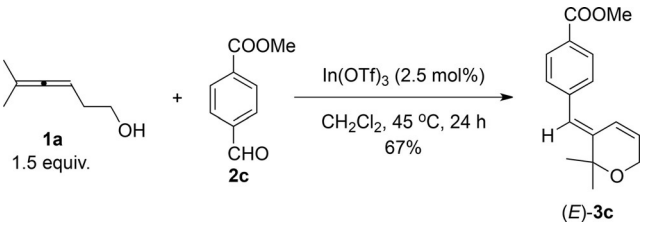
The following compounds were prepared according to this Typical Procedure I.

(*E*)-3-(4-Chlorobenzylidene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3b]:

The reaction of $In(OTf)_3$ (5.8 mg, 0.01 mmol), **2b** (56.2 mg, 0.4 mmol), **1a** (67.8 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3b** (eluent: *n*-hexane/ethyl ether = 40/1) as an oil; yield: 79.9 mg (85%). ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.29$ (d, $J = 8.4$ Hz, 2H, Ar-H), 7.19 (d, $J = 8.4$ Hz, 2H, Ar-H), 6.52 (bd, $J = 10.2$ Hz, 1H, =CH), 6.36 (s, 1H, =CH), 5.96–5.89 (m, 1H, =CH), 4.30 (s, 2H, CH_2), 1.44 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 140.2$, 135.3, 132.5, 130.7, 128.7, 128.2, 121.5, 121.2, 73.6, 61.1, 25.9; MS (EI): *m/z* (%) = 236 [M^+(^{37}Cl), 12.33], 234 [M^+(^{35}Cl), 34.05), 190 (100); IR (neat): $\nu = 3046$, 2975, 2929, 2817, 1487, 1442, 1360, 1258, 1179, 1148, 1122, 1093, 1068, 1011 cm^{-1}; HR-MS (EI): $m/z = 234.0814$, calcd. for $C_{14}H_{15}{}^{35}ClO$ (M^+): 234.0811.

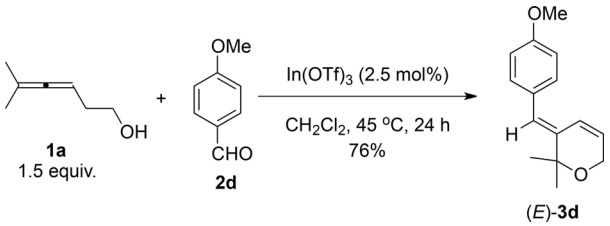


(*E*)-3-(4-Methoxycarbonylbenzylidene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3c]:



The reaction of $In(OTf)_3$ (5.8 mg, 0.01 mmol), **2c** (65.5 mg, 0.4 mmol), **1a** (67.7 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3c** (eluent: *n*-hexane/ethyl ether = 60/1) as a colorless solid; yield: 68.9 mg (67%); mp 109–110 °C (*n*-hexane/ethyl ether). ^1H NMR (400 MHz, $CDCl_3$): $\delta = 8.00$ (d, $J = 8.4$ Hz, 2H, Ar-H), 7.34 (d, $J = 8.4$ Hz, 2H, Ar-H), 6.58–6.54 (m, 1H, =CH), 6.44 (s, 1H, =CH), 6.01–5.95 (m, 1H, =CH), 4.33–4.30 (m, 2H, CH_2), 3.92 (s, 3H, CH_3), 1.46 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 167.1$, 141.9, 141.4, 129.52, 129.49, 129.4, 128.3, 121.7, 121.6, 73.8, 61.2, 52.2, 26.1; MS (EI): *m/z* (%) = 258 (M^+, 83.32), 115 (100); IR (neat): $\nu = 3043$, 2978, 2950, 2928, 2820, 1721, 1606, 1435, 1402, 1379, 1359, 1279, 1221, 1196, 1179, 1149, 1108, 1171, 1018 cm^{-1}; anal. calcd. for $C_{16}H_{18}O_3$: C 74.40, H 7.02; found: C 74.55, H 7.29.

(*E*)-3-(4-Methoxybenzylidene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3d]:

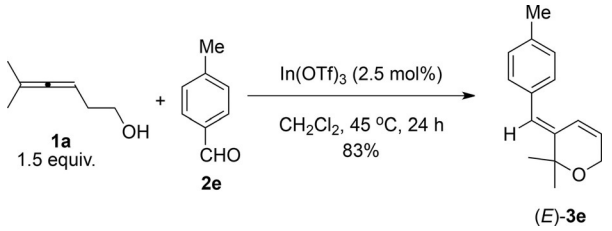


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16154169, 2016, 24, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201600588 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

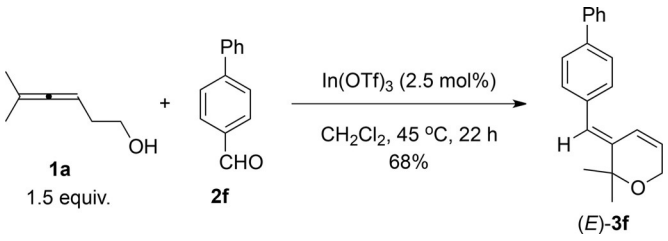
The reaction of $In(OTf)_3$ (5.5 mg, 0.01 mmol), **2d** (54.0 mg, 0.4 mmol), **1a** (68.1 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3d** (eluent: *n*-hexane/ethyl ether = 40/1) as an oil; yield: 69.0 mg (76%). ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.21$ (d, $J = 8.7$ Hz, 2H, Ar-H), 6.87 (d, $J = 8.4$ Hz, 2H, Ar-H), 6.59 (bd, $J = 10.5$ Hz, 1H, =CH), 6.38 (s, 1H, =CH), 5.90–5.85 (m, 1H, =CH), 4.30 (s, 2H, CH_2), 3.80 (s, 3H, CH_3), 1.44 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 158.6$, 138.4, 130.7, 129.5, 127.6, 122.2, 122.1, 113.6, 73.8, 61.2, 55.3, 26.1; MS (EI): *m/z* (%) = 230 (M^+, 35.43), 186 (100); IR (neat): $\nu = 3042$, 2964, 2933, 2821, 1602, 1571, 1508, 1443, 1406, 1374, 1357, 1297, 1247, 1176, 1149, 1121, 1067, 1029, 1010 cm^{-1}; HR-MS (EI): *m/z* = 230.1311, calcd. for $C_{15}H_{18}O_2$ (M^+): 230.1307.

(*E*)-2,2-Dimethyl-3-(4-methylbenzylidene)-3,6-dihydro-2*H*-pyran [(*E*)-3e]:



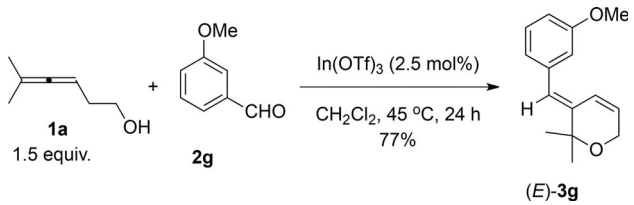
The reaction of $In(OTf)_3$ (6.1 mg, 0.01 mmol), **2e** (47.7 mg, 0.4 mmol), **1a** (67.8 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3e** (eluent: *n*-hexane/ethyl ether = 40/1) as an oil; yield: 70.5 mg (83%). ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.19$–7.11 (m, 4H, Ar-H), 6.60 (bd, $J = 10.2$ Hz, 1H, =CH), 6.40 (s, 1H, =CH), 5.91–5.85 (m, 1H, =CH), 4.31–4.29 (m, 2H, CH_2), 2.34 (s, 3H, CH_3), 1.44 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 139.1$, 136.6, 134.1, 129.4, 128.9, 127.8, 122.6, 122.1, 73.8, 61.2. 26.1, 21.3; MS (EI) *m/z* (%) = 214 (M^+, 40.58), 170 (100); IR (neat): $\nu =$ 3043, 2977, 2924, 2817, 1509, 1448, 1377, 1359, 1260, 1177, 1148, 1122, 1069, 1013 cm^{-1}; HR-MS (EI): *m/z* = 214.1360, calcd. for $C_{15}H_{18}O$ (M^+): 214.1358.

(*E*)-2,2-Dimethyl-3-(4-phenylbenzylidene)-3,6-dihydro-2*H*-pyran [(*E*)-3f]:



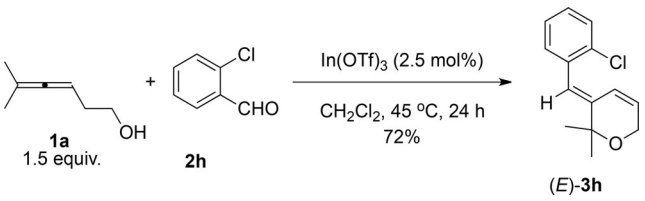
The reaction of $In(OTf)_3$ (5.5 mg, 0.01 mmol), **2f** (72.7 mg, 0.4 mmol), **1a** (67.3 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3f** (eluent: *n*-hexane/ethyl ether = 45/1) as a white solid; yield: 75.0 mg (68%); mp 80–82 °C (*n*-hexane/CH_2Cl_2). ^1H NMR (300 MHz, $CDCl_3$): $\delta = 7.63$–7.54 (m, 4H, Ar-H), 7.43 (t, $J = 7.5$ Hz, 2H, Ar-H), 7.36–7.29 (m, 3H, Ar-H), 6.66 (bd, $J = 10.5$ Hz, 1H, =CH), 6.45 (s, 1H, =CH), 5.96–5.88 (m, 1H, =CH), 4.35–4.25 (m, 2H, CH_2), 1.46 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 140.8$, 139.8, 139.6, 136.1, 130.0, 128.9, 128.3, 127.4, 127.1, 126.9, 122.2, 122.0, 73.8, 61.2, 26.1; MS (EI) *m/z* (%) 276 (M^+, 100); IR (neat): $\nu = 3076$, 3028, 2976, 2925, 2818, 1598, 1486, 1447, 1402, 1378, 1358, 1197, 1178, 1148, 1124, 1071, 1008 cm^{-1}; anal. calcd. for $C_{20}H_{20}O$: C 86.92, H 7.29; found: C 87.02, H 7.30.

(*E*)-3-(3-Methoxybenzylidene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3g]:



The reaction of $In(OTf)_3$ (5.7 mg, 0.01 mmol), **2g** (54.6 mg, 0.4 mmol), **1a** (67.8 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3g** (eluent: *n*-hexane/ethyl ether = 40/1) as an oil; yield: 70.7 mg (77%). ^1H NMR (400 MHz, $CDCl_3$): $\delta = 7.26$–7.21 (m, 1H, Ar-H), 6.86 (d, $J = 7.6$ Hz, 1H, Ar-H), 6.81–6.77 (m, 2H, Ar-H), 6.61 (bd, $J = 10.4$ Hz, 1H, =CH), 6.40 (s, 1H, =CH), 5.92–5.87 (m, 1H, =CH), 4.30–4.29 (m, 2H, CH_2), 3.80 (s, 3H, CH_3), 1.45 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 159.4$, 139.9, 138.4, 129.2, 128.2, 122.5, 122.1, 122.0, 114.9, 112.6, 73.8, 61.2, 55.3, 26.1; MS (EI) *m/z* (%) = 230 (M^+, 100); IR (neat): $\nu = 3044$, 2976, 2933, 2828, 1597, 1576, 1486, 1453, 1430, 1378, 1360, 1316, 1271, 1206, 1155, 1122, 1070, 1044, 1012 cm^{-1}; HR-MS (EI): *m/z* = 230.1308, calcd. for $C_{15}H_{18}O_2$ (M^+): 230.1307.

(*E*)-3-(2-Chlorobenzylidene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3h]:

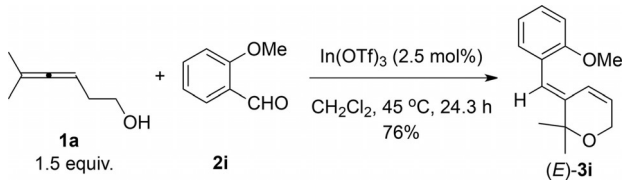


The reaction of $In(OTf)_3$ (5.4 mg, 0.01 mmol), **2h** (56.8 mg, 0.4 mmol), **1a** (66.8 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3h** (eluent: *n*-hexane/ethyl ether = 30/1) as an oil; yield: 68.1 mg (72%). ^1H NMR (400 MHz, $CDCl_3$): $\delta = 7.39$–7.36 (m, 1H, Ar-H), 7.32–7.28 (m, 1H, Ar-H), 7.24–7.17 (m, 2H, Ar-H), 6.46 (s, 1H, =CH), 6.39 (bd, $J = 10.0$ Hz, 1H, =CH), 5.94–5.89 (m, 1H, =CH), 4.33–4.24 (m, 2H, CH_2), 1.48 (s, 6H, $2 \times CH_3$); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 140.7$, 135.1, 134.1, 131.8, 129.4, 128.8, 128.3, 126.2, 121.7, 120.0, 73.9, 61.1, 26.2; MS (EI) *m/z* (%) = 236 [$M^+(^{37}Cl)$, 8.97], 234 [$M^+(^{35}Cl)$, 27.23], 43 (100); IR (neat): $\nu = 3059$, 2998, 2971, 2932, 2877, 2812, 1634, 1587, 1465, 1435, 1408, 1375, 1355, 1329, 1290, 1256, 1236, 1216, 1195, 1177, 1147, 1122, 1069, 1051, 1034, 1011 cm^{-1}; HR-MS (EI): *m/z* = 234.0810, calcd. for $C_{14}H_{15}O^{35}Cl$ (M^+): 234.0811.

(*E*)-3-(2-Methoxybenzylidene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3i]:

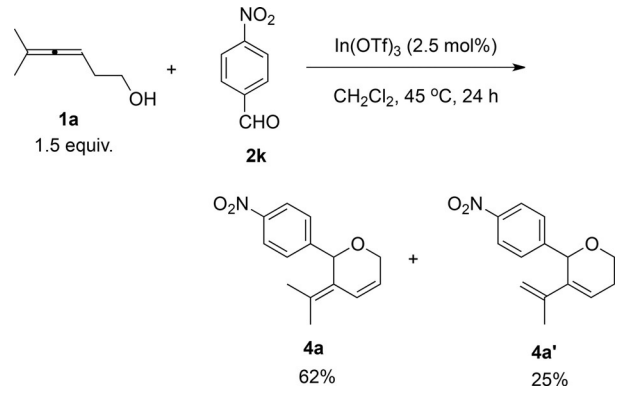
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16154169, 2016, 24, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201600588 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



The reaction of $In(OTf)_3$ (6.0 mg, 0.01 mmol), **2i** (54.3 mg, 0.4 mmol), **1a** (67.4 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded (*E*)-**3i** (eluent: *n*-hexane/ethyl ether = 35/1) as a white solid; 69.5 mg (76%); mp 81–83 °C (*n*-hexane/CH_2Cl_2). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.27–7.20 (m, 2H, Ar-H), 6.92 (t, J = 7.6 Hz, 1H, Ar-H), 6.86 (d, J = 8.0 Hz, 1H, Ar-H), 6.56–6.51 (m, 2H, =CH), 5.89–5.84 (m, 1H, =CH), 4.32–4.27 (m, 2H, CH_2), 3.81 (s, 3H, CH_3), 1.47 (s, 6H, 2 × CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 157.5, 139.1, 131.5, 128.4, 127.4, 125.7, 122.4, 120.0, 118.8, 110.5, 74.0, 61.2, 55.5, 26.2; MS (EI): m/z (%) = 230 (M^+, 51.12), 186 (100); IR (neat): ν = 3016, 2978, 2930, 2871, 2811, 1634, 1593, 1483, 1458, 1431, 1405, 1357, 1297, 1241, 1215, 1192, 1168, 1144, 1108, 1064, 1024 cm^{-1}; anal. calcd. for $C_{15}H_{18}O_2$: C 78.23, H 7.88; found: C 78.26, H 7.76.

2-(4-Nitrophenyl)-3-(propan-2-ylidene)-3,6-dihydro-2*H*-pyran (4a) and 2-(4-nitrophenyl)-3-(propen-2-yl)-5,6-dihydro-2*H*-pyran (4a′):

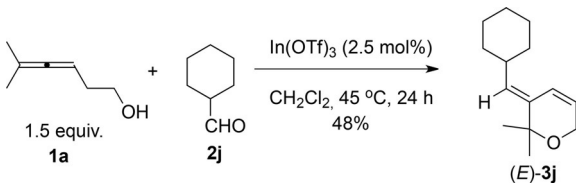


The reaction of $In(OTf)_3$ (5.5 mg, 0.01 mmol), **2k** (60.4 mg, 0.4 mmol), **1a** (67.8 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **4a** (yield: 61.0 mg, 62%) and **4a′** (yield: 24.7 mg, 25%) (eluent: *n*-hexane/ethyl ether = 40/1).

4a: oil; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.20 (d, J = 8.7 Hz, 2H, Ar-H), 7.51 (d, J = 9.0 Hz, 2H, Ar-H), 6.68 (d, J = 10.5 Hz, 1H, =CH), 5.77 (s, 1H, OCH), 5.70 (d, J = 10.2 Hz, 1H, =CH), 4.09 (dd, J = 17.6, 3.5 Hz, 1H, one proton of CH_2), 3.77 (d, J = 17.7 Hz, 1H, one proton of CH_2), 1.96 (s, 3H, CH_3), 1.67 (s, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 147.7, 146.2, 131.2, 129.1, 124.9, 123.9, 123.7, 122.4, 73.5, 60.1, 20.2, 19.6; MS (EI) m/z (%) = 245 (M^+, 35.44), 201 (100); IR (neat): ν = 2957, 2924, 2853, 1739, 1604, 1522, 1459, 1377, 1347, 1260, 1106, 1083, 1016 cm^{-1}; HR-MS (EI): m/z = 245.1048, calcd. for $C_{14}H_{15}NO_3$ (M^+): 245.1052.

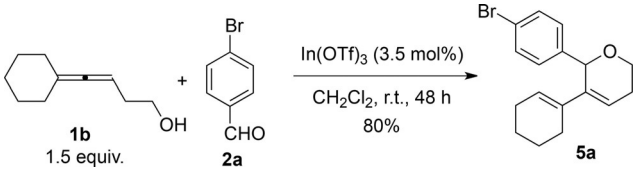
4a′: oil; ^{1}H NMR (300 MHz, $CDCl_3$): δ = 8.19 (d, J = 8.7 Hz, 2H, Ar-H), 7.51 (d, J = 8.7 Hz, 2H, Ar-H), 6.30 (t, J = 4.1 Hz, 1H, =CH), 5.55 (s, 1H, OCH), 4.82 (s, 1H, one proton of =CH_2), 4.49 (s, 1H, one proton of =CH_2), 3.68–3.60 (m, 1H, one proton of CH_2), 3.58–3.39 (m, 1H, one proton of =CH_2), 2.60–2.41 (m, 1H, one proton of CH_2), 2.25–2.18 (m, 1H, one proton of CH_2), 1.91 (s, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 148.1, 147.7, 140.3, 136.6, 130.0, 125.3, 123.6, 113.2, 74.4, 59.1, 25.7, 20.9; MS (EI): m/z (%) = 245 (M^+, 100); IR (neat): ν = 2960, 2918, 2850, 1730, 1606, 1522, 1348, 1262, 1181, 1104, 1089, 1033 1016 cm^{-1}; HR-MS (EI): m/z = 245.1055, calcd. for $C_{14}H_{15}NO_3$ (M^+): 245.1052.

Typical Procedure II: (*E*)-3-(Cyclohexylmethylene)-2,2-dimethyl-3,6-dihydro-2*H*-pyran [(*E*)-3j]:



To a Schlenk tube were added $In(OTf)_3$ (5.5 mg, 0.01 mmol), **2j** (44.9 mg, 0.4 mmol)/2 mL of CH_2Cl_2, **1b** (92.6 mg, 0.6 mmol) and 2 mL of CH_2Cl_2 sequentially. Then the resulting mixture was stirred at room temperature for 24 h as monitored by TLC. Then 10 mL of CH_2Cl_2 and 10 mL of ethyl ether were used to transfer the reaction mixture. After evaporation, 28 μL of mesitylene were added to the residue for the ^{1}H NMR analysis and purified by chromatography on silica gel to afford (*E*)-**3j** (eluent: petroleum ether/ethyl ether = 50/1) as an oil; yield: 39.4 mg (48%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 6.47 (bd, J = 10.4 Hz, 1H, =CH), 5.86–8.76 (m, 1H, =CH), 5.18 (d, J = 8.8 Hz, 1H, =CH), 4.23 (s, 2H, OCH_2), 2.42–2.26 (m, 1H, CH), 1.77–1.55 (m, 5H, five protons of Cy), 1.36–1.00 (m, 11H, 2 × CH_3 + five protons of Cy); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 135.6, 129.7, 129.66, 125.8, 120.96, 120.94, 73.5, 60.7, 35.5, 33.3, 26.08, 26.03, 25.96, 25.88; MS (EI): m/z (%) = 206 (M^+, 66.06), 123 (100); IR (neat): ν = 3043, 2975, 2923, 2850, 1449, 1365, 1264, 1196, 1153, 1103, 1050, 1019 cm^{-1}; HR-MS (EI): m/z = 206.1673, calcd. for $C_{14}H_{22}O$ (M^+): 206.1671.

Typical Procedure III: 2-(4-Bromophenyl)-3-cyclohexenyl-5,6-dihydro-2*H*-pyran (5a):



To a Schlenk tube were added $In(OTf)_3$ (8.2 mg, 0.014 mmol), 4-bromobenzaldehyde **2a** (73.5 mg, 0.4 mmol), 5,5-pentamethylene-3,4-pentadienol **1b** (92.6 mg, 0.6 mmol), and 4 mL of CH_2Cl_2 sequentially. Then the resulting mixture was stirred at room temperature for 48 h as monitored by TLC. Then 10 mL of CH_2Cl_2 and 10 mL of ethyl ether were used to transfer the reaction mixture. After evaporation, 28 μL of mesitylene were added to the residuefor the ^{1}H NMR analysis and purified by chromatography on silica

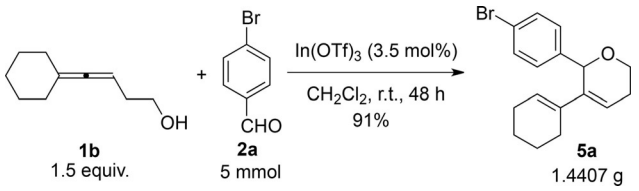
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gel to afford **5a** (eluent: petroleum ether/ethyl ether = 100/1) as white solid; yield: 101.4 mg (80%); mp 63–64 °C (*n*-hexane/ethyl ether). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.46–7.42 (m, 2H, Ar-H), 7.23–7.17 (m, 2H, Ar-H), 6.08 (t, J = 4.2 Hz, 1H, =CH), 5.42–5.38 (m, 1H, OCH), 5.34 (t, J = 3.8 Hz, 1H, =CH), 3.63–3.56 (m, 1H, one proton of OCH_2), 3.53–3.45 (m, 1H, one proton of OCH_2), 2.50–2.37 (m, 1H, one proton of CH_2), 2.24–2.04 (m, 3H, one proton of CH_2 + CH_2), 2.00–1.83 (m, 2H, CH_2), 1.69–1.52 (m, 2H, CH_2), 1.51–1.42 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 139.9, 137.4, 133.7, 131.1, 130.8, 124.6, 121.6, 120.7, 74.5, 58.5, 25.51, 25.46, 22.7, 22.0; MS (EI): m/z (%) = 320 [M^+(^{81}Br], 98.55), 318 [M^+(^{79}Br), 100]; IR (neat): ν = 3045, 2916, 2858, 2718, 2678, 1578, 1480, 1441, 1398, 1336, 1252, 1062, 1006 cm^{-1}; anal. calcd. for $C_{17}H_{19}BrO$: C 63.96, H 6.00; found: C 63.97, H 6.06.

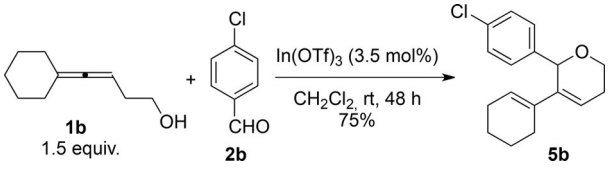
The following compounds were prepared according to this Typical Procedure III.

Synthesis of 2-(4-Bromophenyl)-3-cyclohexenyl-5,6-dihydro-2*H*-pyran (5a) on a 1 g scale:



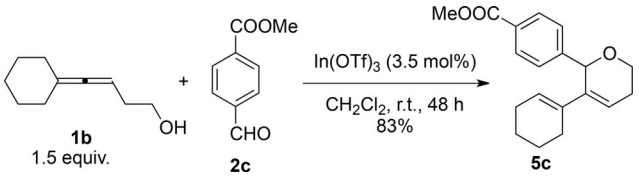
The reaction of $In(OTf)_3$ (0.0992 g, 0.175 mmol), **2a** (0.9203 g, 5 mmol), **1b** (1.1841 g, 7.5 mmol), and CH_2Cl_2 (50 mL) afforded **5a** (eluent: hexane/ethyl ether = 120/1) as white solid; yield: 1.4407 g (91%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.44 (d, J = 8.4 Hz, 2H, Ar-H), 7.20 (d, J = 8.4 Hz, 2H, Ar-H), 6.13–6.02 (m, 1H, =CH), 5.40 (s, 1H, CH), 5.38–5.30 (m, 1H, =CH), 3.66–3.57 (m, 1H, one proton of OCH_2), 3.53–3.45 (m, 1H, one proton of OCH_2), 2.50–2.38 (m, 1H, one proton of CH_2), 2.27–2.04 (m, 3H, one proton of CH_2+CH_2), 2.03–1.84 (m, 2H, CH_2), 1.69–1.41 (m, 4H, 2 × CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 139.9, 137.4, 133.7, 131.1, 130.8, 124.6, 121.6, 120.7, 74.5, 58.5, 25.51, 25.46, 22.7, 22.0.

2-(4-Chlorophenyl)-3-cyclohexenyl-5,6-dihydro-2*H*-pyran (5b):



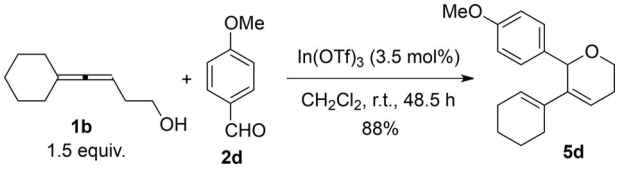
The reaction of $In(OTf)_3$ (8.3 mg, 0.014 mmol), **2b** (56.1 mg, 0.4 mmol), **1b** (90.2 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **5b** (eluent: petroleum ether/ethyl ether = 100/1) as an oil; yield: 82.1 mg (75%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.30–7.24 (m, 4H, Ar-H), 6.08 (t, J = 4.0 Hz, 1H, =CH), 5.42 (s, 1H, OCH), 5.38–5.32 (m, 1H, =CH), 3.63–3.57 (m, 1H, one proton of OCH_2), 3.54–3.46 (m, 1H, one proton of OCH_2), 2.47–2.40 (m, 1H, one proton of CH_2), 2.23–2.04 (m, 3H, one proton of CH_2 + CH_2), 1.99–1.84 (m, 2H, CH_2), 1.69–1.55 (m, 2H, CH_2), 1.52–1.43 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 139.6, 137.7, 133.9, 133.5, 130.6, 128.3, 124.8, 120.9, 74.6, 58.7, 25.7, 25.6, 22.9, 22.2; MS (EI): m/z (%) = 276 [M^+(^{37}Cl), 33.08], 274 [M^+(^{35}Cl), 95.77], 91 (100); IR (neat): ν = 3034, 2941, 2914, 2856, 1627, 1590, 1484, 1436, 1403, 1316, 1281, 1250, 1205, 1175, 1137, 1085, 1055, 1012 cm^{-1}; HR-MS (EI): m/z = 274.1122, calcd. for $C_{17}H_{19}{}^{35}ClO$ (M^+): 274.1124.

3-Cyclohexenyl-2-(4-methoxycarbonylphenyl)-5,6-dihydro-2*H*-pyran (5c):



The reaction of $In(OTf)_3$ (8.2 mg, 0.014 mmol), **2c** (64.4 mg, 0.4 mmol), **1b** (90.5 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **5c** (eluent: petroleum ether/ethyl ether = 80/1 to 40/1) as a white solid; yield: 99.3 mg (83%); mp 109–110 °C (*n*-hexane/ethyl ether). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.00 (d, J = 8.0 Hz, 2H, Ar-H), 7.40 (d, J = 8.0 Hz, 2H, Ar-H), 6.16–6.06 (m, 1H, =CH), 5.48 (s, 1H, OCH), 5.38–5.28 (m, 1H, =CH), 3.90 (s, 3H, OCH3), 3.65–3.60 (m, 1H, one proton of OCH_2), 3.55–3.49 (m, 1H, one proton of OCH_2), 2.47–2.41 (m, 1H, one proton of CH_2), 2.24–2.06 (m, 3H, one proton of CH_2+CH_2), 1.98–1.82 (m, 2H, CH_2), 1.67–1.55 (m, 2H, CH_2), 1.48–1.42 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 167.0, 146.0, 137.4, 133.8, 129.35, 129.33, 129.0, 124.7, 120.8, 74.8, 58.9, 52.0, 25.6, 25.5, 25.4, 22.7, 22.0; MS (EI): m/z (%) = 298 (M^+, 100); IR (neat): ν = 3044, 2971, 2916, 2857, 2834, 1792, 1721, 1608, 1502, 1437, 1410, 1373, 1311, 1273, 1189, 1083, 1019 cm^{-1}; anal. calcd. for $C_{19}H_{22}O_3$: C 76.48, H 7.43; found: C 76.18, H 7.44.

3-Cyclohexenyl-2-(4-methoxyphenyl)-5,6-dihydro-2*H*-pyran (5d):



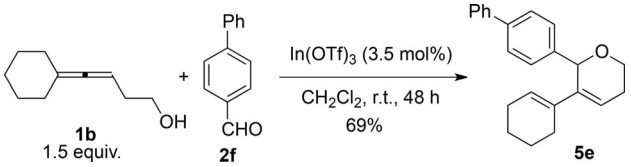
The reaction of $In(OTf)_3$ (8.1 mg, 0.014 mmol), **2d** (54.2 mg, 0.4 mmol), **1b** (91.5 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **5d** (eluent: petroleum ether/ethyl ether = 100/1, then *n*-hexane/ethyl ether = 60/1) as a white solid; yield: 94.9 mg (88%); mp 69–71 °C (*n*-hexane/ethyl ether). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.24 (d, J = 8.8 Hz, 2H, Ar-H), 6.85 (d, J = 8.4 Hz, 2H, Ar-H), 6.06 (t, J = 4.0 Hz, 1H, =CH), 5.49–5.37 (m, 2H, =CH + OCH), 3.80 (s, 3H, OCH_3), 3.62–3.49 (m, 2H, CH_2), 2.50–2.38 (m, 1H, one proton of CH_2), 2.25–2.09 (m, 3H, one proton of CH_2+CH_2), 2.05–1.82 (m, 2H, CH_2), 1.69–1.53 (m, 2H, CH_2), 1.51–1.40 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 159.0, 138.1,

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16154169, 2016, 24, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adsc.201600588 by Zhejiang University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

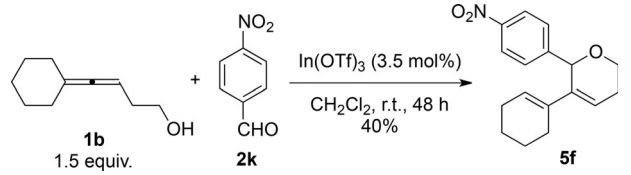
134.0, 133.1, 130.2, 124.4, 120.2, 113.3, 74.7, 58.2, 55.1, 25.58, 25.56, 25.54, 22.8, 22.1; MS (EI): *m/z* (%) = 270 (M^+, 100); IR (neat): ν = 3043, 2995, 2921, 2862, 2838, 1609, 1506, 1452, 1336, 1310, 1243, 1172, 1091, 1047 cm^{-1}; anal. calcd. for $C_{18}H_{22}O_2$: C 79.96, H 8.20; found: C 79.65, H 8.44.

3-Cyclohexenyl-2-(4-phenylbenzylidene)-5,6-dihydro-2*H*-pyran (5e):



The reaction of $In(OTf)_3$ (8.4 mg, 0.014 mmol), **2f** (73.1 mg, 0.4 mmol), **1b** (91.7 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **5e** (eluent: petroleum ether/ethyl ether = 100/1 to 60/1) as a white solid; yield: 87.2 mg (69%); mp 133–135 °C (*n*-hexane/ethyl ether). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.53–7.49 (m, 4H, Ar-H), 7.49–7.25 (m, 5H, Ar-H), 6.09 (s, 1H, =CH), 5.50 (s, 1H, OCH), 5.43 (s, 1H, =CH), 3.71–3.49 (m, 2H, CH_2), 2.52–2.37 (m, 1H, one proton of CH_2), 2.30–2.18 (m, 3H, one proton of CH_2 + CH_2), 2.05–1.85 (m, 2H, CH_2), 1.70–1.40 (m, 4H, 2 × CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 141.0, 140.5, 140.0, 138.0, 134.0, 129.6, 128.8, 127.3, 127.2, 126.9, 124.7, 120.6, 75.0, 58.7, 25.7, 22.9, 22.3; MS (EI): *m/z* (%) = 316 (M^+, 100); IR (neat): ν = 3038, 2916, 2861, 1401, 1335, 1284, 1245, 1212, 1174, 1132, 1061, 1000 cm^{-1}; anal. calcd. for $C_{23}H_{24}O$: C 87.30, H 7.64; Found: C 87.15, H 7.67.

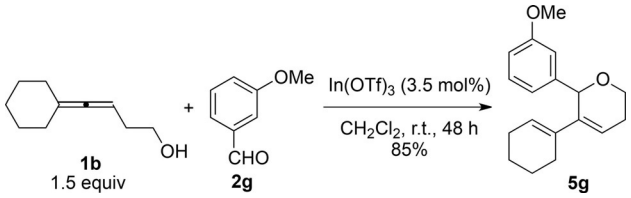
3-Cyclohexenyl-2-(4-nitrophenyl)-5,6-dihydro-2*H*-pyran (5f):



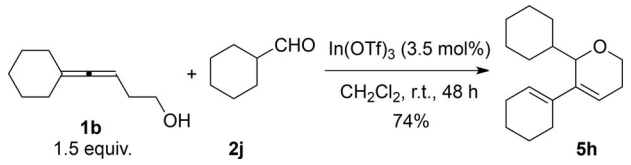
The reaction of $In(OTf)_3$ (8.4 mg, 0.014 mmol), **2k** (60.2 mg, 0.4 mmol), **1b** (91.2 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **5f** (eluent: petroleum ether/ethyl ether = 80/1 to 40/1) as an oil; yield: 45.4 mg (40%). ^1H NMR (300 MHz, $CDCl_3$): δ = 8.19 (d, *J* = 8.1 Hz, 2H, Ar-H), 7.50 (d, *J* = 8.7 Hz, 2H, Ar-H), 6.14 (t, *J* = 4.1 Hz, 1H, =CH), 5.51 (s, 1H, OCH), 5.35–5.27 (m, 1H, =CH), 3.70–3.60 (m, 1H, one proton of CH_2), 3.54–3.43 (m, 1H, one proton of CH_2), 2.54–2.38 (m, 1H, one proton of CH_2), 2.30–1.78 (m, 5H, one proton of CH_2 + 2 × CH_2), 1.74–1.38 (m, 4H, 2 × CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 148.3, 147.3, 136.9, 133.8, 129.8, 124.9, 123.3, 121.3, 74.2, 59.2, 25.6, 25.5, 25.4, 22.6, 22.0; MS (EI): *m/z* (%) = 285 (M^+, 77.27), 91 (100); IR (neat): ν = 3043, 2925, 1601, 1518, 1432, 1343, 1255, 1178, 1137, 1089, 1015 cm^{-1}; HR-MS (EI): *m/z* = 285.1363, calcd. for $C_{17}H_{19}NO_3$ (M^+): 285.1365.

3-(Cyclohexenyl)-2-(3-methoxyphenyl)-5,6-dihydro-2*H*-pyran (5g):

The reaction of $In(OTf)_3$ (8.5 mg, 0.014 mmol), **2g** (55.0 mg, 0.4 mmol), **1b** (90.3 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **5g** (eluent: petroleum ether/ethyl ether = 80/1 to 50/1) as an oil; yield: 93.0 mg (85%). ^1H NMR (400 MHz, $CDCl_3$): δ = 7.23 (t, *J* = 8.4 Hz, 1H, Ar-H), 6.93 (d, *J* = 7.6 Hz, 1H, Ar-H), 6.91–6.87 (m, 1H, Ar-H), 6.84–6.79 (m, 1H, Ar-H), 6.06 (t, *J* = 4.0 Hz, 1H, =CH), 5.42–5.38 (m, 2H, =CH + OCH), 3.79 (s, 3H, OCH_3), 3.63–3.53 (m, 2H, CH_2), 2.47–2.41 (m, 1H, one proton of CH_2), 2.25–2.06 (m, 3H, one proton of CH_2 + CH_2), 1.99–1.86 (m, 2H, CH_2), 1.68–1.54 (m, 2H, CH_2), 1.50–1.43 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 159.3, 142.4, 137.8, 133.9, 128.9, 124.3, 121.5, 120.3, 114.7, 112.9, 75.1, 58.5, 55.1, 25.53, 25.50, 22.7, 22.1; MS (EI): *m/z* (%) = 270 (M^+, 100); IR (neat): ν = 3041, 2999, 2926, 2860, 2832, 1599, 1585, 1485, 1463, 1451, 1435, 1319, 1259, 1156, 1091, 1050 cm^{-1}; HR-MS (EI): *m/z* = 270.1622, calcd. for $C_{18}H_{22}O_2$ (M^+): 270.1620.

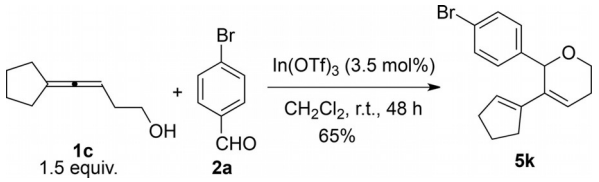


3-(Cyclohexenyl)-2-cyclohexyl-5,6-dihydro-2*H*-pyran (5h):



The reaction of $In(OTf)_3$ (19.3 mg, 0.035 mmol), **2j** (112.8 mg, 1 mmol), **1b** (220.2 mg, 1.5 mmol), and CH_2Cl_2 (10 mL) afforded **5h** (eluent: petroleum ether/ethyl ether = 120/1) as an oil; yield: 183.7 mg (74%). ^1H NMR (300 MHz, $CDCl_3$): δ = 5.80 (d, *J* = 6.0 Hz, 1H, OCH), 5.57–5.48 (m, 1H, =CH), 4.30–4.22 (m, 1H, =CH), 3.96–3.88 (m, 1H, one proton of CH_2), 3.47 (td, *J* = 10.5, 3.4 Hz, 1H, one proton of CH_2), 2.33–2.14 (m, 2H, CH_2), 2.11–2.02 (m, 2H, CH_2), 1.98–1.84 (m, 2H, CH_2), 1.82–1.34 (m, 11H, one proton of Cy + 5 × CH_2), 1.31–1.00 (m, 4H, 2 × CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 142.0, 136.7, 122.5, 120.5, 78.3, 62.8, 40.9, 30.2, 27.2, 27.0, 26.6, 26.4, 25.6, 25.5, 25.3, 22.9, 22.3; MS (EI): *m/z* (%) = 246 (M^+, 15.95), 163 (100); IR (neat): ν = 2922, 2850, 1448, 1380, 1346, 1334, 1302, 1262, 1235, 1217, 1196, 1177, 1114, 1094, 1039 cm^{-1}; HR-MS (EI): *m/z* = 246.1979, calcd. for $C_{17}H_{26}O$ (M^+): 246.1984.

2-(4-Bromophenyl)-3-(cyclopentenyl)-5,6-dihydro-2*H*-pyran (5k):

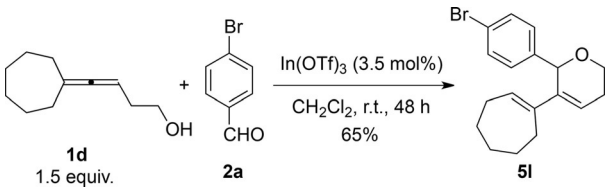


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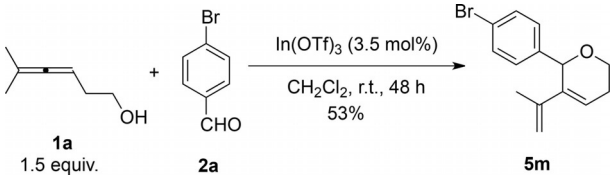
The reaction of $In(OTf)_3$ (20.1 mg, 0.035 mmol), **2a** (184.7 mg, 1 mmol), **1c** (206.8 mg, 1.5 mmol), and CH_2Cl_2 (10 mL) afforded **5k** (eluent: petroleum ether/ethyl ether = 100/1) as a white solid; yield: 197.7 mg (65%); m.p. 70–71 °C (hexane/ethyl ether). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.44 (d, J = 8.3 Hz, 2H, Ar-H), 7.21 (d, J = 8.1 Hz, 2H, Ar-H), 6.08–6.00 (m, 1H, =CH), 5.38 (s, 1H, OCH), 5.14 (s, 1H, =CH), 3.66–3.50 (m, 2H, OCH_2), 2.58–2.38 (m, 3H, one proton of CH_2 + CH_2), 2.34–2.11 (m, 3H, one proton of CH_2+CH_2), 1.87–1.74 (m, 2H, CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 140.3, 139.9, 133.8, 131.2, 130.8, 126.8, 123.7, 121.7, 75.0, 58.6, 33.3, 32.2, 25.5, 22.3; MS (EI): m/z (%) = 306 [M^+(^{81}Br), 53.47], 304 [M^+(^{79}Br), 53.69], 91 (100); IR (neat): ν = 3047, 2856, 1897, 1581, 1478, 1404, 1304, 1253, 1194, 1070, 1004 cm^{-1}; anal. calcd. for $C_{16}H_{17}BrO$: C 62.96, H 5.61; found: C 63.03, H 5.71.

2-(4-Bromophenyl)-3-(cycloheptenyl)-5,6-dihydro-2*H*-pyran (5l):



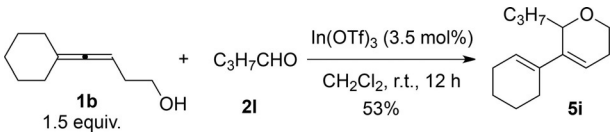
The reaction of $In(OTf)_3$ (19.2 mg, 0.035 mmol), **2a** (184.1 mg, 1 mmol), **1d** (247.6 mg, 1.5 mmol), and CH_2Cl_2 (10 mL) afforded **5l** (eluent: petroleum ether/ethyl ether = 80/1) as an oil; yield: 215.5 mg (65%). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.40 (d, J = 8.1 Hz, 2H, Ar-H), 7.19 (d, J = 8.0 Hz, 2H, Ar-H), 6.06 (t, J = 4.0 Hz, 1H, =CH), 5.57 (t, J = 6.9 Hz, 1H, =CH), 5.38 (s, 1H, OCH), 3.68–3.53 (m, 2H, CH_2), 2.46–2.14 (m, 4H, 2 × CH_2), 2.08–1.94 (m, 2H, CH_2), 1.73–1.56 (m, 2H, CH_2), 1.46–1.20 (m, 4H, 2 × CH_2); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 142.3, 139.7, 139.2, 131.1, 130.8, 129.0, 121.6, 121.2, 75.2, 59.3, 32.3, 29.6, 28.2, 26.4, 26.1, 25.5; MS (EI): m/z (%) = 334 [M^+(^{81}Br), 52.33], 332 [M^+(^{79}Br), 53.74], 91 (100); IR (neat): ν = 3042, 2917, 2848, 1642, 1589, 1574, 1484, 1443, 1403, 1340, 1325, 1285, 1252, 1211, 1177, 1089, 1070, 1053, 1011 cm^{-1}; HR-MS (EI): m/z = 332.0778, calcd. for $C_{18}H_{21}{}^{79}BrO$ (M^+): 332.0776.

2-(4-Bromophenyl)-3-(propen-2-yl)-5,6-dihydro-2*H*-pyran (5m):



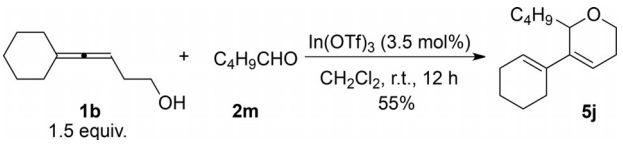
The reaction of $In(OTf)_3$ (20.2 mg, 0.035 mmol), **2a** (184.4 mg, 1 mmol), **1a** (169.2 mg, 1.5 mmol), and CH_2Cl_2 (10 mL) afforded **5m** (eluent: petroleum ether/ethyl ether = 80/1) as a white solid; yield: 148.1 mg (53%); mp 82–85 °C (hexane/ethyl ether). ^{1}H NMR (400 MHz,$CDCl_3$): δ = 7.45 (d, J = 8.1 Hz, 2H, Ar-H), 7.20 (d, J = 8.4 Hz, 2H, Ar-H), 6.23 (t, J = 3.6 Hz, 1H, =CH), 5.43 (s, 1H, OCH), 4.80 (s, 1H, one proton of =CH_2), 4.53 (s, 1H, one proton of =CH_2), 3.66–3.58 (m, 1H, one proton of OCH_2), 3.56–3.48 (m, 1H, one proton of OCH_2), 2.54–2.41 (m, 1H, one proton of CH_2), 2.25–2.14 (m, 1H, one proton of CH_2), 1.89 (s, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 140.2, 139.6, 137.0, 131.3, 130.7, 124.5, 121.8, 112.9, 74.5, 58.3, 25.6, 20.7; MS (EI): m/z (%) = 280 [M^+(^{81}Br), 35.48], 278 [M^+(^{79}Br), 34.90], 79 (100); IR (neat): ν = 3087, 3046, 2977, 2950, 2902, 2862, 2726, 2694, 1901, 1766, 1633, 1606, 1588, 1481, 1462, 1441, 1401, 1376, 1346, 1284, 1253, 1216, 1173, 1106, 1086, 1067, 1030, 1009 cm^{-1}; anal. calcd. for $C_{14}H_{15}{}^{79}BrO$: C 60.23, H 5.42; found: C 60.11, H 5.53.

Typical Procedure IV: 3-(Cyclohexenyl)-2-*n*-propyl-5,6-dihydro-2*H*-pyran (5i):



To a Schlenk tube were added $In(OTf)_3$ (8.0 mg, 0.014 mmol), **2l** (28.8 mg, 0.4 mmol)/2 mL of CH_2Cl_2, **1b** (92.6 mg, 0.6 mmol)/2 mL of CH_2Cl_2 sequentially. Then the resulting mixture was stirred at room temperature for 12 h as monitored by TLC. Then 10 mL of CH_2Cl_2 and 10 mL of ethyl ether were used to transfer the reaction mixture. After evaporation, 28 μL of mesitylene were added to the residue for the ^{1}H NMR analysis and purified by chromatography on silica gel to afforded **5i** (eluent: petroleum ether/ethyl ether = 150/1) as an oil; yield: 43.0 mg (53%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 5.74 (t, J = 4.0 Hz, 1H, OCH), 5.56 (t, J = 3.8 Hz, 1H, =CH), 4.43–4.37 (m, 1H, =CH), 3.87–3.79 (m, 1H, one proton of CH_2), 3.64–3.56 (m, 1H, one proton of CH_2), 2.25–2.05 (m, 5H, 2 × CH_2+one proton of CH_2), 2.00–1.89 (m, 1H, one proton of CH_2), 1.71–1.34 (m, 8H, 4 × CH_2), 0.92 (t, J = 7.2 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 142.1, 135.6, 122.6, 118.6, 73.3, 59.8, 35.0, 26.6, 25.55, 25.49, 22.8, 22.3, 18.9, 14.0; MS (EI): m/z (%) = 206 (M^+, 31.13), 163 (100); IR (neat): ν = 2923, 2862, 1451, 1377, 1337, 1264, 1104, 1033 cm^{-1}; HR-MS (EI): m/z = 206.1669, calcd. for $C_{14}H_{22}O$ (M^+): 206.1671.

3-(Cyclohexenyl)-2-*n*-butyl-5,6-dihydro-2*H*-pyran (5j):



Following Typical Procedure IV, the reaction of $In(OTf)_3$ (8.1 mg, 0.014 mmol), **2m** (34.1 mg, 0.4 mmol), **1b** (91.5 mg, 0.6 mmol), and CH_2Cl_2 (4 mL) afforded **5j** (eluent: petroleum ether/ethyl ether = 150/1) as an oil; yield: 48.4 mg (55%). ^{1}H NMR (400 MHz, $CDCl_3$): δ = 5.74 (t, J = 4.0 Hz, 1H, OCH), 5.56 (t, J = 3.8 Hz, 1H, =CH), 4.42–4.35 (m, 1H, =CH), 3.88–3.79 (m, 1H, one proton of CH_2), 3.65–3.56 (m, 1H, one proton of CH_2), 2.25–2.04 (m, 5H, 2×CH_2+one proton of CH_2), 2.00–1.89 (m, 1H, one proton of CH_2), 1.72–1.15 (m, 10H, 5 × CH_2), 0.90 (t, J = 7.0 Hz, 3H, CH_3);

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^{13}C NMR (100 MHz, $CDCl_3$): δ = 142.1, 135.6, 122.6, 118.6, 73.5, 59.9, 32.6, 28.0, 26.6, 25.5, 25.48, 22.8, 22.6, 22.3, 14.1; MS (EI): *m/z* (%) = 220 (M^+, 27.32), 163 (100); IR (neat): ν = 3034, 2924, 2856, 1450, 1377, 1337, 1265, 1104, 1044 cm^{-1}; HR-MS (EI): *m/z* = 220.1821, calcd. for $C_{15}H_{24}O$ (M^+): 220.1827.

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